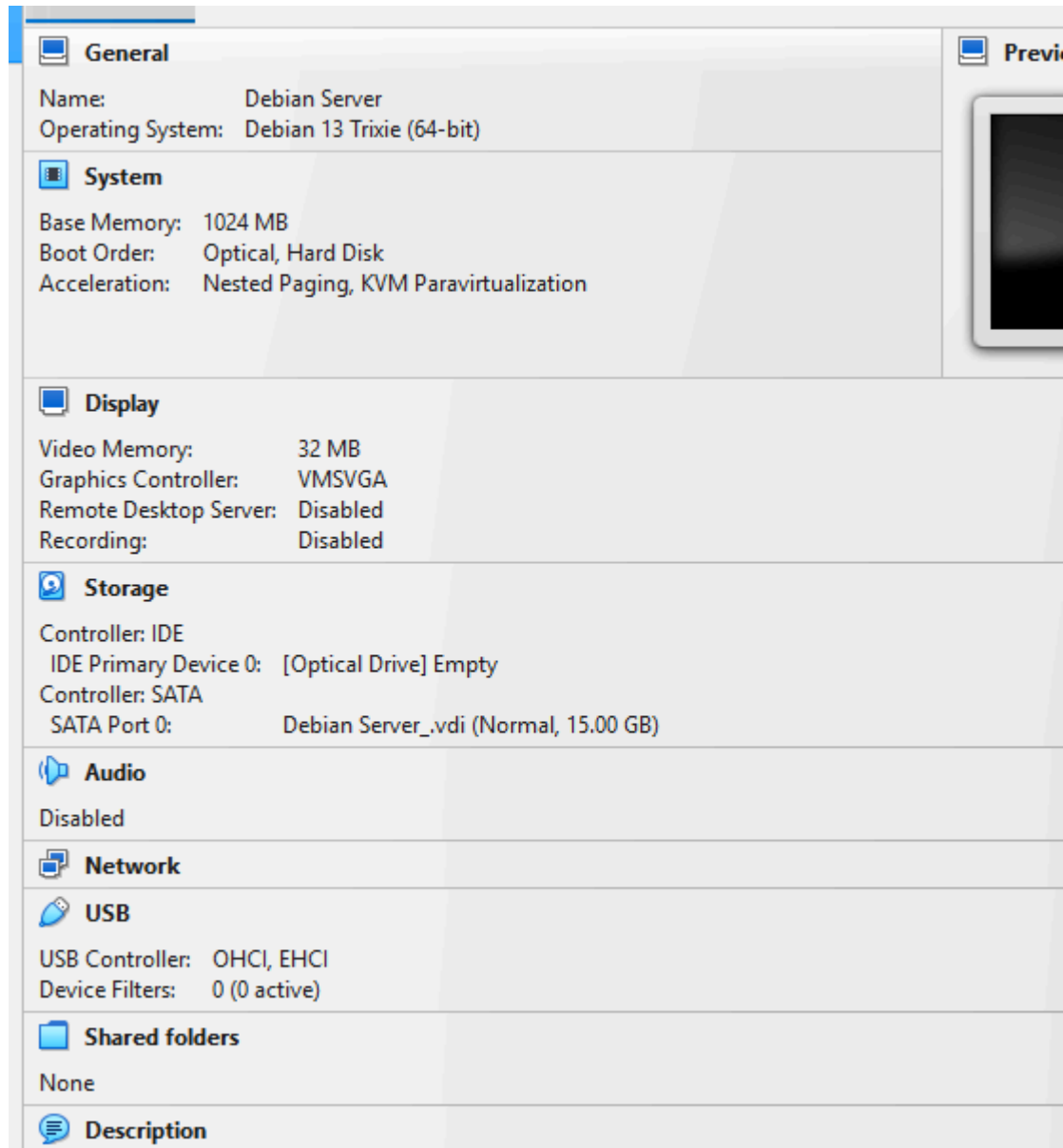


Part 2 of Project

What are the server hardware specifications?



What is the Debian Login Screen?

```
Debian GNU/Linux 13 webserver tty1
webserver login: luella
Password:
Linux webserver 6.12.85+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.85-1 (2026-04-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
luella@webserver:~$
```

What is the IP address of your Debian Server VM?

```
luella@webserver:~$ hostname -I
10.0.2.15 fd17:625c:f037:2:7098:3b17:e289:c687
luella@webserver:~$ _
```

How do you work with the Firewall in Debian?

How do you check if the Firewall is running?

To check if the firewall is running, you can run the command: `sudo ufw status`

How do you disable the Firewall?

To disable the firewall, you can run the command `sudo ufw disable`

How do you add Apache to the Firewall?

To add Apache to the firewall you must run the following two commands `sudo ufw allow 'WWW Full'`
`sudo ufw allow 'OpenSSH'`

What different commands do we use to work with Apache?

`systemctl status apache2`

This command is used to check if Apache is enabled and running.

```

jella@webserver:~$ systemctl status apache2
apache2.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
  Active: active (running) since Wed 2026-05-06 01:50:13 EDT; 27min ago
  Invocation: d05ed581bece4b5fac2b23aaa68988a5
  Docs: https://httpd.apache.org/docs/2.4/
  Main PID: 697 (apache2)
  Tasks: 55 (limit: 1093)
  Memory: 8.4M (peak: 8.6M)
  CPU: 128ms
  CGroup: /system.slice/apache2.service
          └─697 /usr/sbin/apache2 -k start
            └─698 /usr/sbin/apache2 -k start
              └─700 /usr/sbin/apache2 -k start

Warning: some journal files were not opened due to insufficient permissions.
jella@webserver:~$

```

`sudo systemctl start apache2`

This command is used to start the web server.

`sudo systemctl stop apache 2`

This command is used to stop the web server.

`sudo systemctl restart apache2`

This command stops and then restarts the web server.

`sudo apachectl configtest`

This command is used to test the Apache configuration. ~[ApacheTestConfig](#)

`sudo apache2 -v`

This command is used to check the version of Apache.

What are some common configuration files for Apache?

Common configuration files for Apache appear to be

- `htaccess` (per-directory configuration file)
- `httpd.conf`: (primary configuration file for Apache web server)
- `apache2.conf`: (same purpose as `httpd.conf` but for Debian conventions)

```

jella@webserver:~$ sudo ls /etc/apache2
apache2.conf  conf-available  conf-enabled  envvars  magic  mods-available  mods-enabled  ports.conf  sites-available  sit
jella@webserver:~$ _

```

Where does Apache store logs?

Apache stores logs in `/var/log/apache2/access.log` - by Default Apache also stores error logs in `/var/log/apache2/error.log`

```
uella@webserver:~$ sudo ls /etc/apache2
apache2.conf  conf-available  conf-enabled  envvars  magic  mods-available  mods-enabled  ports.conf  sites-available  sit
uella@webserver:~$ sudo ls /var/log/apache2/access.log
/var/log/apache2/access.log
uella@webserver:~$
```

What are some commands we can use to review logs?

The easiest command to view our logs is `sudo cat /var/log/apache2/access.log` and you can replace `access.log` with `error.log` for the error logs.

```
uella@webserver:~$ sudo ls /etc/apache2
apache2.conf  conf-available  conf-enabled  envvars  magic  mods-available  mods-enabled  ports.conf  sites-available  sit
uella@webserver:~$ sudo ls /var/log/apache2/access.log
/var/log/apache2/access.log
uella@webserver:~$ sudo cat /var/log/apache2/access.log
1 - - [06/May/2026:01:26:25 -0400] "GET / HTTP/1.1" 200 11014 "-" "wget/1.25.0"
1 - - [06/May/2026:01:26:33 -0400] "GET / HTTP/1.1" 200 11014 "-" "wget/1.25.0"
0.0.2.2 - - [06/May/2026:01:33:49 -0400] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWe
147.0.0.0 Safari/537.36"
0.0.2.2 - - [06/May/2026:01:33:49 -0400] "GET /icons/openlogo-75.png HTTP/1.1" 200 6040 "http://127.0.0.1:8080/" "Mozilla
WebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0.0.2.2 - - [06/May/2026:01:33:49 -0400] "GET /favicon.ico HTTP/1.1" 404 529 "http://127.0.0.1:8080/" "Mozilla/5.0 (Windo
.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
0.0.2.2 - - [06/May/2026:01:34:40 -0400] "-" 408 0 "-" "-"
0.0.2.2 - - [06/May/2026:01:51:58 -0400] "GET / HTTP/1.1" 200 3383 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWe
147.0.0.0 Safari/537.36"
0.0.2.2 - - [06/May/2026:01:52:50 -0400] "-" 408 0 "-" "-"
uella@webserver:~$ _
```

You can use `sudo tail /var/log/apache2/access.log` If you wish to view the most recent entries in the log files.
